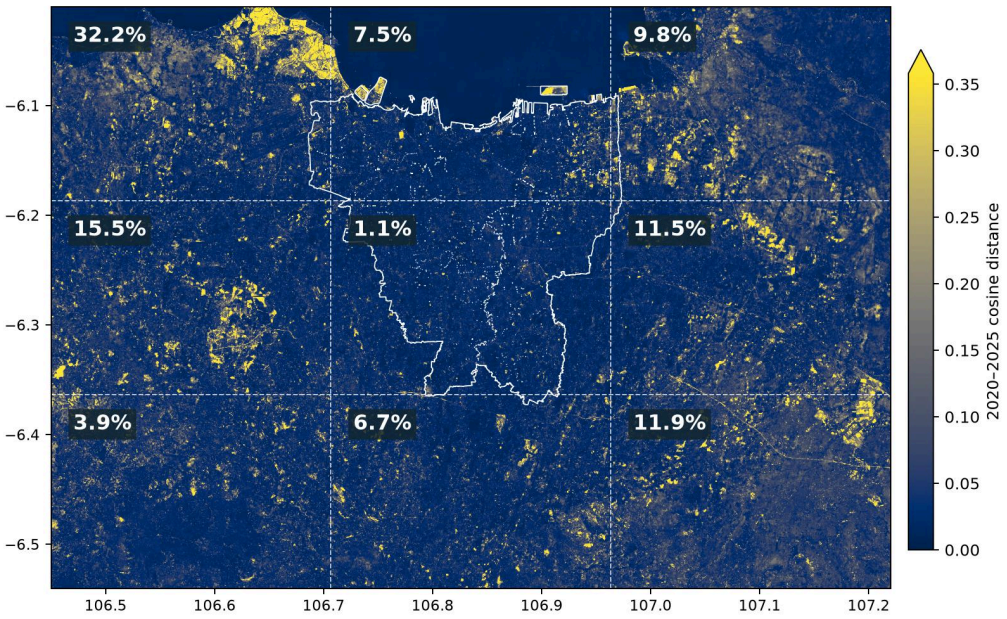
GEOSPATIAL CASE STUDY / NABHONIL SARKAR / NOVEMBER 2026

Satellite-observed surface differences around Jakarta, 2020–2025

I started with a question about subsidence risk: does sinking ground have the same implications beneath homes, factories and open land? My radar measurements could not support a reliable absolute rate. I narrowed the aim to a question the observations can answer: where and when did satellite observations of the land surface around Jakarta differ most? I used Google's AlphaEarth embeddings—annual 64-number descriptions of the land surface—sampled onto a common nominal 30 m grid.

I SET THREE OBJECTIVES:

1. reproduce the annual processing on one grid;
2. compare the timing and location of the largest differences;
3. and test how much independent evidence supports their interpretation.



Mainland DKI outlined in white. Higher values indicate less similar embeddings.

Brighter colours mean less similar satellite observations, not faster sinking. Labels give each sector's share of all top-1% cells: those in the highest 1% of 2020–2025 differences. A valid cell is a location with usable observations in both compared years. The white outline marks mainland Jakarta.

The northwest of the study area, mostly west of mainland Jakarta's boundary, contained 32.2% of the top-1% cells but only 11.1% of all valid cells.

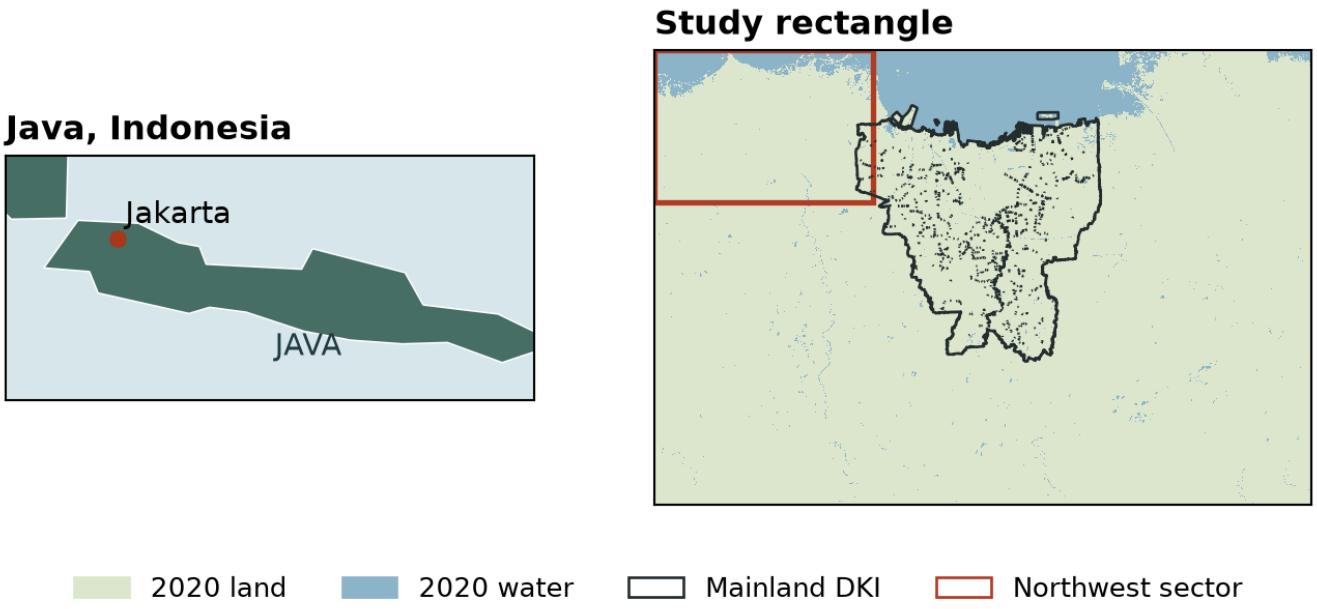
The northwest still stood out when I restricted the comparison to land mapped in 2020 and when I excluded the narrow western strip supplied by a separate source tile. That makes it a place to investigate, not evidence of reclamation.

01

STUDY AREA AND TOOLS

What was measured and where

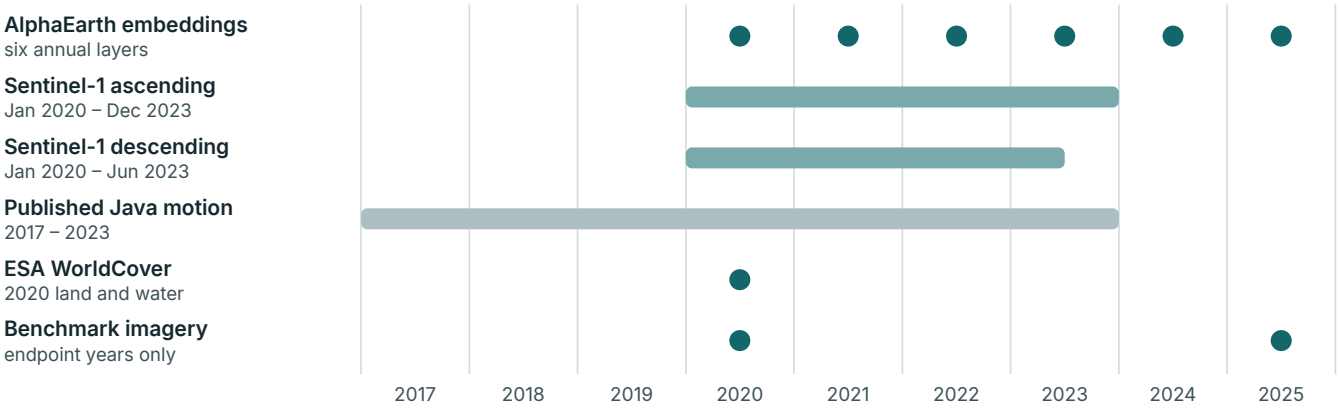
The study rectangle on Java's north coast spans approximately 85 × 59 km (106.45–107.22°E, 6.54–6.01°S). It includes sea and areas outside the administrative city. "Northwest" means the upper-left ninth of that rectangle, where 96.8% of valid cells and 98.8% of top-1% cells lie outside mainland Jakarta; "North Jakarta" means the administrative area used in the radar coverage check.



Regional inset: Natural Earth land. Local view: WorldCover 2020 at grid centres, not a 2025 shoreline. The rectangular frame is the analysis extent.

TASK	DATA OR TOOL	WHAT IT CONTRIBUTED
Ground movement	*COMET–LiCSAR, LiCSBAS, Colab*	Sentinel-1 radar pairs and time-series quality checks; line-of-sight motion relative to a reference.
Annual surface context	*AlphaEarth via Google Cloud Storage*	Six annual 64-number embeddings per location. Earth Engine supplied the earlier 2023 reference export only.
Alignment and comparison	*Python: Rasterio, NumPy, GeoPandas*	Sample grid centres, compare all axes, screen missing cells and align geographic boundaries.
Interpretation	*ESA WorldCover; dated imagery; published Java observations*	Check land-only sensitivity, audit independent imagery and retain measured ground-motion points separately.
Software verification	*pytest; Git and GitHub Actions*	Local regression tests, reproducible code revisions and automated repository checks. 183 tests passed locally with project data; GitHub Actions also runs automated checks.

DATA COVERAGE



Dots mark annual products or imagery endpoints. Bars show the observation period covered by each source.

02

FROM THE ORIGINAL QUESTION TO THE CURRENT METHOD

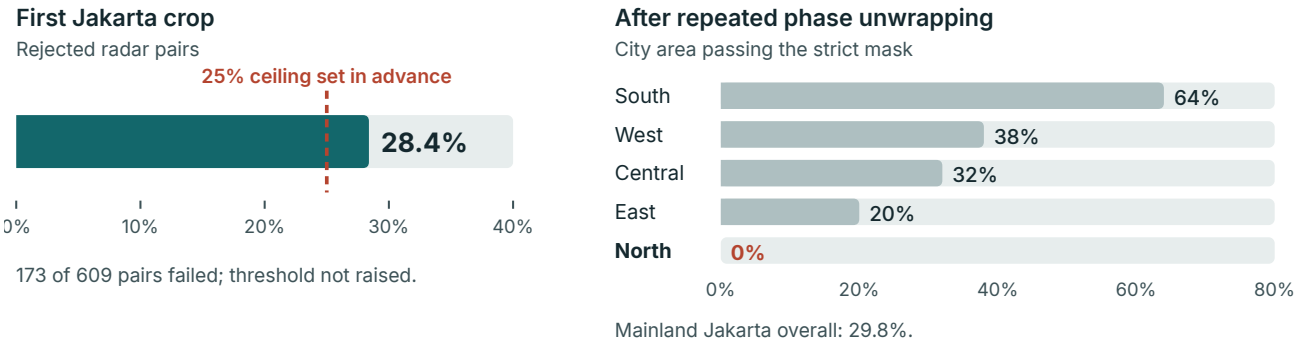
Why I withdrew the absolute subsidence rate

Radar measures motion along the satellite's line of sight. I kept those physical measurements separate from the 64-number embeddings: similarity in land appearance is not a measure of subsidence, exposed assets or vulnerability.

The ascending track lost coverage in North Jakarta

I recovered 609 complete radar pairs. Full-frame runs that averaged pixels by factors of 2 and 5 (multilooking) failed loop-closure checks, which test whether phase changes around a closed loop of dates add up to near zero; the coarser factor-10 result remained diagnostic only. Cropping to Jakarta improved network survival, but 173 of 609 pairs still failed the loop-closure check, above the 25% limit I had set for that experiment. The checks retained LiCSBAS's option for accounting for overall loop-phase bias. I did not raise the threshold. After I repeated phase unwrapping consistently for all 609 radar pairs, the retained network passed its connectivity checks, but only 29.8% of mainland Jakarta passed the strict mask; no area in North Jakarta passed.

ASCENDING-TRACK QUALITY CHECKS



Left: first Jakarta crop; the 25% ceiling applied to this experiment, not as a universal rejection standard. Right: after repeated phase unwrapping, by city.

A relative pattern survived on a different track

On the descending track (January 2020–June 2023), the strict mask covered 68.4% of North Jakarta. Changing the reference area preserved the spatial pattern after removing the overall reference offset, but shifted the median line-of-sight rate across a range of 8.05 mm/year. My limit was 3 mm/year. I kept the pattern as a case study and withdrew the absolute-rate claim. I did not combine the tracks or fill uncovered areas.

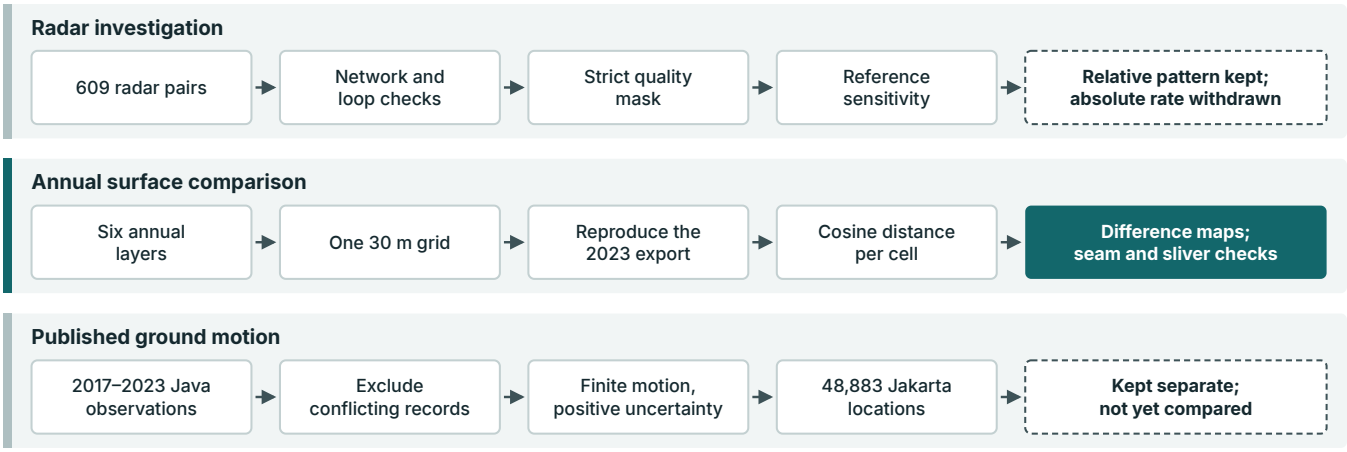
A software check changed a label

A synthetic −150 mm/year rate fell outside the original classification ranges and silently received the default label "stable." I changed the code to require an explicit class for every valid value. An unclassified value now stops the pipeline.

03 How the annual comparisons were made

METHOD AND QUALITY CHECKS

HOW THE EVIDENCE IS ORGANISED



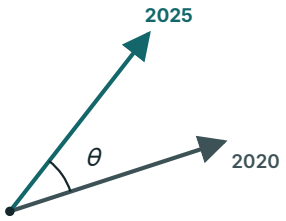
Three lines of evidence. Each ends in its own output; none feeds another.

Reproduce one year before comparing six

Sampling Google's native 10 m product at 18 reference locations reproduced none of the earlier 2023 export vectors. Sampling its stored 20 m overview at the centres of the frozen nominal 30 m grid reproduced every one of the 64 values at all 5,619,719 in-boundary cells. I used that same method for 2020 through 2025. This checks my processing, not the stability of the source product across years.

Measure difference without assigning a cause

I decoded all 64 axes, normalised each vector and calculated one minus the cosine of the angle between years. A larger value means less similar observations. I used jointly valid cells, including water; I did not independently standardise each year. The endpoint compares 2020 and 2025 directly. It is not the sum of the five annual changes.



distance = 1 − cos θ

same direction → 0; wider angle → larger

Each year's 64 values form a direction. Only the angle between years is compared.

Test the tile seam before interpreting the northwest

The two source tiles met at the same line—the seam—in all six years despite different identifiers. Before examining strip contrasts, I fixed two primary tests: the 2022–2023 jump and the 2020–2025 endpoint, both on all valid cells. I compared 120 m strips on each side of the line along 28 two-kilometre segments, against 20 parallel control lines. An unusual step had to be stronger than every eligible control and consistent in direction. Appendix A gives the full rule.

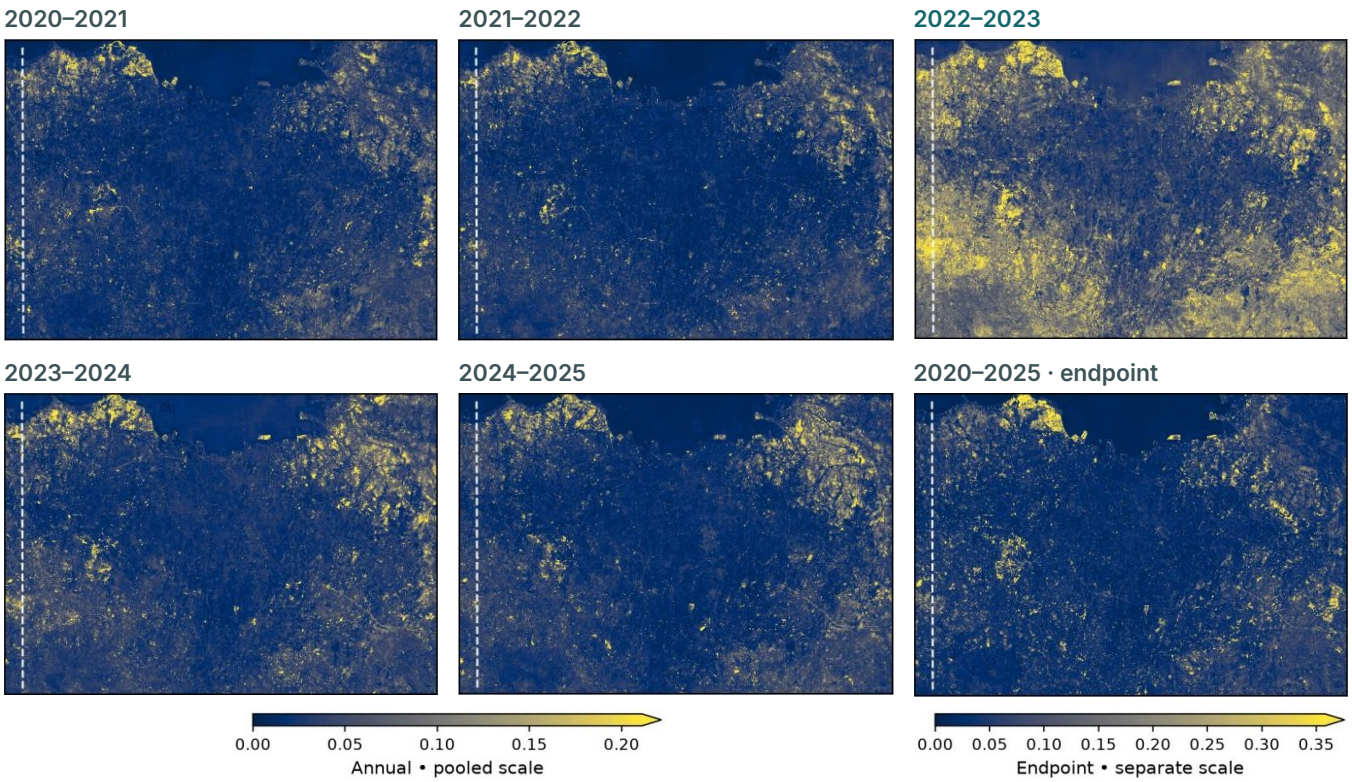
A separate check removed the whole western sliver—the strip west of the seam—across the study rectangle. I kept the original northwest boundary and compared top-1% cells both at the original cutoff and after recalculating the 99th-percentile cutoff.

Screen published ground motion independently

The published 2017–2023 Java observations have different dates and measurement support. Repeated records conflicted at 40,661 coordinates; I excluded those locations rather than averaging them. Requiring finite vertical motion and positive reported uncertainty left 48,883 mainland Jakarta locations, including 10,051 in North Jakarta. These counts describe observations, not continuous coverage.

04
RESULTS BY YEAR

The largest annual difference came in 2022 to 2023



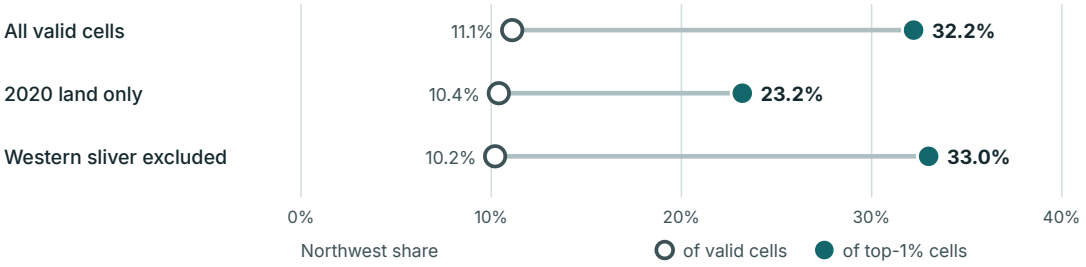
Dashed line: tested tile seam. At or above the shared annual colour cap (0.21112), the shares of valid cells were 2020–21: 0.52%, 2021–22: 0.55%, 2022–23: 2.21%, 2023–24: 1.01%, 2024–25: 0.71%. Endpoint: 1.00% at its separate 0.35809 cap. Brighter capped cells cannot be distinguished by colour. Statistics use uncapped values.

Brightening in 2022–2023 appears broad, particularly in outer and southern sectors, and less pronounced centrally. The median was 0.112 in the southeast and 0.033 in the north-central sector. This describes geography; it does not establish a cause.

05

RESULTS BY PLACE AND TIME

The northwest concentration survives both exclusions



Northwest share (%) • open: valid cells / filled: top-1% cells

On land mapped in 2020, the concentration weakens but persists: the northwest held 23.2% of the land-only top-1% cells on 10.4% of valid land cells. This is a conservative coastal check: it also excludes any land created after 2020 in cells mapped as water that year.

Only 254 of the northwest's 18,096 original top-1% cells lay west of the seam (1.4%). Removing the western sliver left 32.97% of all remaining top-1% cells in the northwest at the old cutoff; recalculating the cutoff gave 32.99%, on 10.20% of remaining valid cells. The headline does not depend on that sliver.

Neither primary comparison flagged the seam

In 2022–2023 the seam contrast was weaker than all 20 control lines, with only 50% directional agreement. For the endpoint it exceeded 11 of 20, with 57% agreement. Of 18 tests, 12 did not flag and six water-only tests were inconclusive: only two segments qualified, against six required. This argues against an unusually sharp step at this width, not against a shift shared by both tiles.

The jump was widespread but did not persist everywhere

For 85.4% of valid cells, 2022–2023 was the largest of the five annual changes. Its median distance was 0.069, compared with 0.024–0.032 for the other transitions. In 2024, 81% of cells were closer to their 2023 than their 2022 vector. In 2025, 68% were closer to 2022. That reversal concerns the observations; it does not mean the land reverted. The pattern points to a possible influence shared by 2023 and 2024; temporary surface or observation conditions and product-related differences both remain possible.

I also computed a direct 2020–2023 comparison, which is not mapped here. It ends in the year of the jump. The direct 2020–2025 map may be less affected by that particular shift; this does not show which map is more accurate.

06 What the evidence supports

OBJECTIVES AND CONCLUSION

OBJECTIVE	OUTCOME	EVIDENCE AND LIMITS
1 **Reproduce the annual processing on one grid**	• Met for processing	The six years use one sampling rule. The 2023 reference was reproduced across all valid cells; this is not an across-year accuracy test.
2a **Compare the timing and location of the largest differences**	• Described	The northwest held the largest share of 2020–2025 top-1% cells. The southeast had the highest sector median for the 2022–2023 transition. These answer different questions.
2b **Robustness of the northwest result**	• Completed with limits	The concentration persisted without the western sliver and, more weakly, on 2020 land. The water-only seam tests lacked support.
3a **Test independent evidence: dated imagery**	• Not established	Benchmark not scored. Independent dated imagery was insufficient for the pre-set spatial benchmark against vegetation and built-up indices.
3b **Published ground motion: contextual comparison**	• Not yet compared	Screened and kept separate. Spatial units and minimum point support are not yet defined.
Original risk-zoning ambition	• Withdrawn as a deliverable	Separate from the three objectives. No validated absolute deformation field or adequate independent exposure evidence.

Why the benchmark stopped

I selected 300 patches without using their change scores and kept spatial blocks separate for development and evaluation. Across 93 releases of Esri's World Imagery Wayback archive, 94 patches had candidate imagery at both endpoints. The development split covered 16 blocks and evaluation covered 22; both needed 20. Pléiades catalogue footprints did not provide access to full-resolution imagery. I therefore did not score embeddings against vegetation and built-up indices.

What I would test next

I would compare the 2022–2023 jump across surface types and check dated imagery at locations selected independently of the change scores. That is a proposed test, not a result. A northwest site selected from this map could illustrate a high score, but cannot independently validate the method.

Before comparing published subsidence with surface differences, I would define spatial units, minimum point support and sensitivity to exclusions. For now, the deliverable is a reproducible map of differences and two findings worth checking: the northwest concentration and the widespread 2022–2023 jump. It is not a hazard, exposure or risk classification.

Explore the analysis and tests on GitHub ↗

APPENDIX A
READING AND REPRODUCING THE MAPS

Definitions and reproducibility

TERM	MEANING HERE
Embedding	A 64-number description of a grid cell for one year; not a land-use label.
Cosine distance	Difference in vector direction. A colour cap controls display only; the 99th percentile is not a noise floor.
Top-1% cell	A cell in the highest 1% of differences for one comparison. The 99th-percentile cutoff is the threshold that defines them.
Valid cell	An in-boundary grid centre with finite, non-zero embedding vectors in both years.
Line of sight	Motion toward or away from the satellite relative to a reference; not automatically vertical subsidence.
Seam	The shared source-tile edge at easting 663,840 m in UTM zone 48S. The analysis grid itself is geographic (EPSG:4326).

Frozen seam test

> Use the median east-minus-west contrast per segment; strength is the absolute median across matched segments. Require ≥60 cells on each side, ≥6 matched segments and ≥19 controls (≥9 per side). A flag must exceed all controls in strength and meet both 75% sign consistency and the control 90th percentile of consistency. Controls are not independent trials. All five annual pairs and the endpoint were tested for all cells, land and water.

Reproduction and sources

From the repository root and locked environment, run the following modules in order with `python -m scripts.<module>`: `build_alphaearth_full_grid`, `audit_alphaearth_full_2023`, `alphaearth_full_grid_change`, `render_alphaearth_change`. Large local rasters are not shipped with the repository. The archived `insar-risk-zoning` command is not the annual workflow.

The registered seam test ran once at commit `ed87fd936daacb7609981296dfcd87612e1d3067`. All 183 tests passed locally on Windows with project data before the run. The changelog records the protocol and result hashes, all 18 outcomes and the original counts.

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5. Zanaga, D., et al. (2021). ESA WorldCover 10 m 2020 v100. Zenodo. doi.org/10.5281/zenodo.5571936

Repository and commands ↗ Test record ↗ AlphaEarth source guide ↗
WorldCover 2020 ↗ Published Java observations ↗ Natural Earth land ↗

The AlphaEarth Foundations Satellite Embedding dataset is produced by Google and Google DeepMind. Land mask: © ESA WorldCover project 2020, v100 (CC BY 4.0). Regional inset: Natural Earth, public domain.